JIAYAN (STEPHEN) XIE

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EDUCATION

Duke Kunshan University **Kunshan, China**

B.S. in Molecular Bioscience / Cell and Molecular Biology *Aug 2023 – May 2027 (Expected)*

Cumulative GPA: 3.4/4.0 | Junior-Year GPA: 3.8/4.0

Duke University **Durham, NC, USA**

B.S. in Interdisciplinary Studies (Molecular Bioscience / Cell and Molecular Biology), Dual Degree Program

Honors & Awards: Dean's List, Duke University (Spring 2026) — Semester GPA 3.925; iGEM Silver Medal (Oct 2024), Team Leader; DKU Undergraduate Scholarship, Duke Kunshan University (Aug 2023) — 170,000 RMB

RESEARCH EXPERIENCE

TUBGCP2 Functional Study in Prostate Cancer Cells

Huang Lab, DKU | Supervisor: Prof. Linfeng Huang | Signature Work Independent Project *Mar 2026 – Present*

Overview: This project uses siRNA knockdown in prostate cancer cells to assess whether TUBGCP2 is a cancer dependency gene, with BIRC5 as a benchmarking control to distinguish biological effects from technical artifacts.

- Designed three independent siRNAs targeting TUBGCP2 and BIRC5 in PC-3 and DU145 cells; systematically evaluated knockdown efficiency via qPCR and Western blot
- Established a multi-layered control framework: FAM-siRNA (delivery control), GAPDH-siRNA (knockdown workflow control), KIF11-siRNA (phenotypic positive control), and NC-siRNA (negative baseline) for systematic troubleshooting of the experimental system
- Planned functional assays upon confirmation of knockdown, including CCK-8 cell viability, cell proliferation counting, and morphological imaging
- Current status: Optimizing qPCR conditions and transfection efficiency; KIF11 positive control shows reduced cell count and morphological abnormalities, while TUBGCP2 knockdown groups have not yet exhibited obvious phenotypes

Functional Genomics in Insect Cell Models to Study Cellular Pathways

Huang Lab, DKU | Supervisor: Prof. Linfeng Huang | Summer Research Scholars Program *Jun 2026 – Oct 2026*

Overview: This project extends pro-siRNA technology to insect cell models (S2, Sf9/Sf21, C6/36) and combines it with high-content imaging to establish a phenotypic screening platform for identifying genes regulating cell viability, RNA delivery, and gene silencing.

- Contributed to preparing over 30 96-well pro-siRNA library plates; performed high-content imaging-based quantification of dead/live cell ratios at 72 hours post-transfection, identifying multiple candidate wells with altered cell viability
- Developed an eGFP fluorescent reporter system; compared PEI and Lipofectamine 3000 transfection reagents; achieved ~30% transfection efficiency with Lipofectamine 3000; detected sequence-specific gene silencing via co-transfection of eGFP mRNA with eGFP-targeting siRNA
- Established a complete workflow for quantifying cell viability changes, RNA delivery efficiency, and gene silencing, successfully extending pro-siRNA technology from mammalian cancer cells to insect cell models

Construction and Verification of Long Hairpin RNA Plasmids for Recombinant siRNA Production in E. coli

Huang Lab, DKU | Supervisor: Prof. Linfeng Huang | Summer Research Scholars Program *Apr 2025 – Nov 2025*

Overview: This project uses E. coli to produce recombinant siRNA via long hairpin RNA plasmids, providing a more efficient and cost-effective platform for high-throughput RNAi screening.

- Constructed long hairpin RNA plasmids for 17 cancer-related genes by selecting 200–689 bp regions downstream of ATG start codons and performing two-step sequential cloning (SacI/XhoI + NotI/SalI)
- Completed full molecular cloning workflow: high-fidelity PCR, restriction digestion, ligation, transformation, colony screening, plasmid extraction, and Sanger sequencing
- Optimized digestion conditions (extended incubation from 1.5 to 2 hours, doubled enzyme and DNA input), improving cloning efficiency and reproducibility

- Detected ~21 nt siRNA products via RNA gel electrophoresis with consistent 260 nm UV absorption peaks; completed validation for all 17 constructs

Tampon X: Engineered Lactic Acid Bacteria-Based System for Prevention of Tampon-Associated Toxic Shock Syndrome *Dec 2023 – Oct 2024*

DKU BSL-2 Laboratory, DKU iGEM Team | Nisin Wet Lab Group Leader | Supervisor: Prof. Linfeng Huang

Overview: This project engineers Lactococcus lactis (nisin enhancement) and Lactobacillus johnsonii (H_2O_2 enhancement) to inhibit Staphylococcus aureus and prevent Toxic Shock Syndrome, exploring microbiome-based preventive strategies.

- Led the Nisin group and directed the promoter screening module: tested 14 lactic acid bacterial promoters; identified PnisA and PTSIIC-celA as strong expression promoters via ELISA and Western blot
- Constructed pMG36e-AcGFP reporter vectors; validated that NisRK overexpression enhances PnisA-driven expression by increasing GFP signal
- Contributed to CRISPR/Cas9 genome editing strategy design for replacing the native PnisR promoter with a strong promoter to boost nisin production
- Team achievements: Completed six DBTL iteration cycles; constructed and verified an ~12 kb pMG-Cas vector via PCR and single-enzyme digestion; H_2O_2 route achieved a peak of 41.2 μM under anaerobic conditions with LJ_0548/LJ_0549 overexpression; awarded iGEM Silver Medal 2024

SELECTED ACADEMIC PROJECTS

Micro/Nanoplastic Neurotoxicity Mechanism Study *Feb 2025 – Apr 2025*

- Systematically reviewed the molecular pathways by which micro/nanoplastics translocate from the gut to the bloodstream, cross the blood-brain barrier, and induce mitochondrial dysfunction
- Constructed a causal framework linking environmental exposure (ROS overproduction, ATP depletion) to neurodegenerative phenotypes
- Compared size-dependent effects (micro-scale vs. nano-scale) on biological barrier penetration and cytotoxicity, identifying nanoplastics as posing a more direct threat at the organelle level due to their ability to enter mitochondria

AI-Powered Multi-Modal Clinical Decision Support System *Oct 2025 – Nov 2025*

- Designed a conceptual framework for an AI-based diagnostic tool integrating genetic, imaging, and clinical data to address pharmacogenetic challenges, such as CYP2D6-guided analgesia and DPYD-guided fluoropyrimidine dosing
- Conducted pharmacogenetics background research across osteoarthritis, cardiovascular, and oncology contexts; evaluated cross-disease applicability
- Proposed an iterative model training framework incorporating standardized clinical workflows and AI optimization through successive prediction–adjustment cycles; provided preliminary assessment of commercialization potential

EXTRACURRICULAR ACTIVITIES

Varsity Ultimate Frisbee Team

Spirit Captain *Jan 2025 – Present*

- Led pre-tournament rules briefings and coordinated in-game communication among teammates and opponents to clarify calls and resolve disputes
- Facilitated post-match debriefs and reinforced fair play and sportsmanship, earning multiple Best Spirit Awards

SKILLS

Laboratory Skills: *Mammalian and insect cell culture, siRNA transfection and knockdown, RNA immunoprecipitation (RIP), qPCR primer design and validation, gradient PCR optimization, agarose and RNA gel electrophoresis, RT-qPCR, Western blot (SDS-PAGE, membrane transfer, and immunodetection), ELISA, molecular cloning (PCR, restriction digestion, ligation, transformation, plasmid extraction), Sanger sequencing analysis, high-content imaging, flow cytometry, CCK-8 cell viability assay*

Software: *RStudio, GraphPad Prism, Primer3*

Languages: *Chinese (Native), English (Fluent)*